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Module 1 Linear Algebra I

Core concepts

1. Determinant
2. Rank of a matrix, solving a system
3. Characteristic equation, eigen values, eigen vectors.
4. Cayley Hamilton theorem
5. Quadratic form, Reduction to canonical form

Determinant

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $|A| = ad - bc$

If $|A| \neq 0 \Rightarrow A$ is non-singular

1. Find $|A|$, if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$|A| = 4 - 6 = \underline{\underline{-2}}$$

2. Check whether $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 2 & 1 \\ 2 & 4 & 6 \end{bmatrix}$ singular or not

$$|A| = 1(12 - 4) - 2(24 - 2) + 3(16 - 4)$$

$$= 12 - 4 - 48 + 4 + 48 - 12 = 0$$

$\therefore A$ is singular

Rank of a matrix

It is the order of largest square sub matrix whose determinant is non zero.

Q. Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & 4 & 6 \end{bmatrix}$

Ans Step 1 $\Rightarrow |A| = 0$ [row 1 $\times$ 3 = row 2]

Step 2 :- consider sub matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}, \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} 3 & 6 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} 3 & 9 \\ 2 & 6 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}, \begin{bmatrix} 6 & 9 \\ 4 & 6 \end{bmatrix}$$

All submatrices of order 2 have determinant 0.

Step 3 :- considering matrix of order 1

$$[1], [2], \dots$$

$$[1] = 1 \neq 0$$

$$\therefore \text{Rank} = \text{order of } [1] = 1$$

Row reduced Echelon form

Rank $\rightarrow$ The number of non-zero rows in row reduced echelon form.

Allowed Row operations $* R_i \rightarrow R_i \pm k R_j$

$$* R_i \rightarrow k R_i$$

$$* R_i \leftrightarrow R_j \rightarrow R_i \text{ interchanged with } R_j$$

wrong or not allowed

$$* R_i \rightarrow R_i \pm k$$

Q. Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & 4 & 6 \end{bmatrix}$ using row reduced echelon form

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \\ 2 & 4 & 6 \end{bmatrix} \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \text{ change it into } \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{matrix}$$

No. of non-zero rows = 1

$$\therefore \text{Rank of } A = \rho(A) = 1$$

Q. Find the rank of $D = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$$\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \quad D \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -3 & -3 \\ 0 & -2 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 / -3 \quad D \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & -2 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2 \quad D \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Rank of } D = \underline{\underline{\rho(D) = 3}}$$

Solve $x + 3y + 2z = 0$
 $2x + 3y + z = 0$
 $x + y + z = 0$

Matrix Equ

consider $Ax = B$ where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$[A:B]$
 $[A:B] = \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 2 & 3 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 3 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$

$P(A:B) = 3$ Rank of $[A:B] = \text{Rank of } A$

$P(A) = 3 \rightarrow$ System is consistent and has unique solution

Equivalent Equations

$x + 3y + 2z = 0$

$0x + y + z = 0$

$0x + 0y + z = 0$

Applying Back Substitution

$z = 0$

$y + z = 0$

$y = 0 - z - 0 = 0$

$x = -2z - 3y = 0$

Solve $2x + 3y + 4z = 31$

$x - 2y + 4z = 4$

$3x + y - 3z = 2$

$A \rightarrow$ coefficient matrix

$Ax = B$

where $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & -2 & 4 \\ 3 & 1 & -3 \end{bmatrix}$, $x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} 31 \\ 4 \\ 2 \end{bmatrix}$

$[A:B]$
 $[A:B] = \left[\begin{array}{ccc|c} 2 & 3 & 4 & 31 \\ 1 & -2 & 4 & 4 \\ 3 & 1 & -3 & 2 \end{array} \right]$ $R_1 \leftrightarrow R_2$

$\sim \left[\begin{array}{ccc|c} 1 & -2 & 4 & 4 \\ 2 & 3 & 4 & 31 \\ 3 & 1 & -3 & 2 \end{array} \right]$

$\sim \left[\begin{array}{ccc|c} 1 & -2 & 4 & 4 \\ 0 & 7 & -4 & 23 \\ 0 & 7 & -15 & -10 \end{array} \right]$

$R_2 \rightarrow R_2 - 2R_1$

$R_3 \rightarrow R_3 - 3R_1$

$\sim \left[\begin{array}{ccc|c} 1 & -2 & 4 & 4 \\ 0 & 1 & -\frac{4}{7} & \frac{23}{7} \\ 0 & 7 & -15 & -10 \end{array} \right]$ $R_2 \rightarrow R_2 / 7$

$\sim \left[\begin{array}{ccc|c} 1 & -2 & 4 & 4 \\ 0 & 1 & -\frac{4}{7} & \frac{23}{7} \\ 0 & 0 & -11 & -33 \end{array} \right]$ $R_3 \rightarrow R_3 - 7R_2$

$\sim \left[\begin{array}{ccc|c} 1 & -2 & 4 & 4 \\ 0 & 1 & -\frac{4}{7} & \frac{23}{7} \\ 0 & 0 & 1 & 3 \end{array} \right]$ $R_3 \rightarrow R_3 / -11$

$$P(A:B) = P(A) = 3 \Rightarrow \text{no. of variables}$$

$\therefore$ System is consistent and has unique solution

Equivalent equation

$$x - 2y + 4z = 4$$

$$y - \frac{4}{3}z = \frac{23}{3}$$

$$z = \underline{\underline{3}}$$

$$y = \frac{23}{3} + \frac{12}{3} = \frac{35}{3} = \underline{\underline{5}}$$

$$x - 2 \times 5 + 4 \times 3 = 4$$

$$x - 10 + 12 = 4$$

$$\underline{\underline{x = 2}}$$

Suppose A is a square matrix of order n

$I \rightarrow$ identity matrix

* Characteristic matrix of A is $A - \lambda I$

$$\text{eg } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \text{ ch matrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{bmatrix}$$

* Characteristic Polynomial $|A - \lambda I| = A$ polynomial of degree n in λ

* Characteristic equation of A is $|A - \lambda I| = 0$

* Eigen values / characteristic roots are root of equation $|A - \lambda I| = 0$

* Eigen vectors, corresponding to λ are the vectors X that satisfies $[A - \lambda I]X = 0$

1. Find the characteristic equation, eigen values and eigen vectors of $A = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$

$$\text{Ans. Ch. eqn is } |A - \lambda I| = 0$$

$$\text{is } \begin{vmatrix} 1-\lambda & -2 \\ 1 & 4-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(4-\lambda) + 2 = 0$$

$$4 - 5\lambda + \lambda^2 + 2 = 0$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\text{Eigen values are } (\lambda - 3)(\lambda - 2) = 0$$

$$\underline{\underline{\lambda_1 = 3, \lambda_2 = 2}}$$

Eigen vector corresponding to eigen values $\lambda = 2$

$$[A - 2I]X = 0$$

$$\begin{bmatrix} -1 & -2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x - 2y = 0$$

$$x + 2y = 0$$

$$\text{Assign } y = k$$

$$x = -2k$$

$$X = \begin{bmatrix} -2k \\ k \end{bmatrix}$$

$$\text{When } k = 1$$

$$X = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\text{no. of variables} = 2$$

$$\text{rank} = 1$$

Eigen Vector corresponding to $\lambda = 3$

$$[A - 3I]X = 0$$

$$\begin{bmatrix} -2 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2x - 2y = 0$$

$$x + y = 0$$

$$x = -y$$

$$\text{When } x = c$$

$$y = -c$$

$$X = \begin{bmatrix} c \\ -c \end{bmatrix} \quad \text{When } c = 1, \text{ then } X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Cayley Hamilton Theorem

Every square matrix A satisfies its characteristic equation.

$$\text{eg: } A = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} \quad \text{ch. eqn of } A \text{ is } \lambda^2 - 5\lambda + 6 = 0$$

$$\text{CHT means } A^2 - 5A + 6I = 0$$

1 a) Verify CHT for $A = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$

b) Find A^{-1} using CHT

c) Find A^2 using CHT

$$\text{Ans ch. eqn of } A \text{ is } (\lambda - 3)(\lambda - 2) = 0 \Rightarrow \lambda^2 - 5\lambda + 6 = 0$$

$$A^2 = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -10 \\ 5 & 14 \end{bmatrix}$$

$$5A = \begin{bmatrix} 5 & -10 \\ 5 & 20 \end{bmatrix}$$

$$6I = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

$$A^2 - 5A + 6I = \begin{bmatrix} -1 & -10 \\ 5 & 14 \end{bmatrix} - \begin{bmatrix} 5 & -10 \\ 5 & 20 \end{bmatrix} + \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

b) $A^2 - 5A + 6I = 0$

$$A^2 A^{-1} - 5A A^{-1} + 6I A^{-1} = 0 A^{-1}$$

$$A - 5I + 6A^{-1} = 0$$

$$6A^{-1} = 5I - A$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{6} \end{bmatrix}$$

c) $A^2 - 5A + 6I = 0$

$$A^2 = 5A - 6I = \begin{bmatrix} 5 & -10 \\ 5 & 20 \end{bmatrix} - \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -10 \\ 5 & 14 \end{bmatrix}$$

Module 1

Short cuts

1. To find characteristic equation of 2×2 matrix A
 $\Rightarrow \lambda^2 - (\text{Trace } A)\lambda + |A| = 0$ eg: $A = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$, $|A - \lambda I| = 0 \Rightarrow \lambda^2 - 5\lambda + 6 = 0$

3×3 matrix A , $\Rightarrow \lambda^3 - (\text{Trace } A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

2. To find eigen vectors corresponding to

$a_1x + b_1y + c_1z = 0$ Here A_{11} is minor of a_{11}

$a_2x + b_2y + c_2z = 0$

$a_3x + b_3y + c_3z = 0$

Write coefficients of 2 different equations (not constant multiplication)

$$\begin{array}{ccc} a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{array} \begin{array}{c} x \\ y \\ z \end{array} \begin{array}{ccc} a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{array} \Rightarrow \frac{x}{\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}} = \frac{y}{\begin{vmatrix} c_2 & a_2 \\ c_3 & a_3 \end{vmatrix}} = \frac{z}{\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}} = k$$

1. Find the eigen values and eigen vectors of

a $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

$|A - \lambda I| = 0$

$\lambda^3 - (\text{Trace } A)\lambda^2 + (A_{11} + A_{22} + A_{33})\lambda - |A| = 0$

$\text{Trace } A = 8 + 7 + 3 = 18$

$A_{11} = \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} = 21 - 16 = 5$

$A_{22} = \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} = 24 - 4 = 20$

$A_{33} = \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} = 56 - 36 = 20$

$|A| = 8(21 - 16) + 6(-18 + 8) + 2(24 - 14)$
 $= 8 \times 5 + 6 \times -10 + 2 \times 10$
 $= 40 - 60 + 20 = 0$

$\therefore \lambda^3 - 18\lambda^2 + (5 + 20 + 20)\lambda - 0 = 0$

$\lambda^3 - 18\lambda^2 + 45\lambda = 0$

$\lambda(\lambda^2 - 18\lambda + 45) = 0$

$\lambda = 0, (\lambda - 3)(\lambda - 15) = 0$

$\lambda = 0, 3, 15$

$\lambda = 0, \lambda = 3, \lambda = 15$

Eigen vector corresponding to $\lambda = 0$

$[A - 0I]x = 0$

$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$8x - 6y + 2z = 0$

$-6x + 7y - 4z = 0$

$2x - 4y + 3z = 0$

$\begin{array}{ccc} x & y & z \\ 8 & -6 & 2 \\ -6 & 7 & -4 \end{array} \begin{array}{c} x \\ y \\ z \end{array} \begin{array}{ccc} 8 & -6 & 2 \\ -6 & 7 & -4 \end{array}$

$\frac{x}{24 - 14} = \frac{y}{-12 + 32} = \frac{z}{56 - 36} = k$

$\frac{x}{10} = \frac{y}{20} = \frac{z}{20} = k \therefore x = 10k, y = 20k, z = 20k$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 15$

$$[A - 15I]X = 0$$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-7x - 6y + 2z = 0$$

$$-6x - 8y - 4z = 0$$

$$2x - 4y - 12z = 0$$

$$\begin{array}{ccccccc} & x & y & z & & & \\ -7 & -6 & 2 & -7 & -6 & & \\ -6 & -8 & -4 & -6 & -8 & & \end{array}$$

$$\frac{x}{24+16} = \frac{y}{-12-28} = \frac{z}{56-36} = K$$

$$\frac{x}{40} = \frac{y}{-40} = \frac{z}{20} = K$$

$$x = 40K, y = -40K, z = 20K$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = 3$

$$[A - 3I]X = 0$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$5x - 6y + 2z = 0$$

$$-6x + 4y - 4z = 0$$

$$2x - 4y = 0$$

$$\begin{array}{ccccccc} & x & y & z & & & \\ 5 & -6 & 2 & 5 & -6 & & \\ -6 & 4 & -4 & -6 & 4 & & \end{array}$$

$$\frac{x}{24-8} = \frac{y}{-12+20} = \frac{z}{20-36} = K$$

$$\frac{x}{16} = \frac{y}{8} = \frac{z}{-16} = K$$

$$x = 16K, y = 8K, z = -16K$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

b. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ $|A| = -2(-12) - 2(-6) - 3(-4+1) = 24 + 12 + 9 = 45$

$$|A - \lambda I| = 0$$

$$\lambda^3 + \lambda^2 + (-12-3-6)\lambda + 45 = 0$$

$$\lambda^3 + \lambda^2 - 21\lambda + 45 = 0$$

$$\lambda = 5, -3$$

Eigen vector corresponding to $\lambda = 5$

$$[A - 5I]X = 0$$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$-7x + 2y - 3z = 0$$

$$2x - 4y - 6z = 0$$

$$-x - 2y - 5z = 0$$

$$\begin{array}{ccccc} & & x & y & z \\ -7 & -7 & 2 & -3 & -7 & 2 \\ & 2 & -4 & -6 & 2 & -4 \end{array}$$

$$\frac{x}{-12-12} = \frac{y}{-6-42} = \frac{z}{28-4} = k$$

$$\frac{x}{-24} = \frac{y}{-48} = \frac{z}{24} = k$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = -3$

$$(A + 3I) = 0.$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} x + 2y - 3z = 0 \\ 2x + 4y - 6z = 0 \\ -x - 2y + 3z = 0 \end{array}$$

$$\begin{array}{ccccc} & x & y & z & \\ 1 & 2 & -3 & 1 & 2 \\ 2 & 4 & -6 & 2 & 4 \end{array}$$

Here, all equations are constant multiplication.

In that case, put $z = c, y = k$

$$x = 3c - 2k$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3c - 2k \\ y \\ c \end{bmatrix}$$

Assignment Topic Module 1

1. What is Quadratic form.

Quadratic form is a homogeneous polynomial of degree 2 in any no. of variable

$$\text{eg: } 1. x^2 + 3xy + y^2 = [x \ y] \begin{bmatrix} 1 & 3/2 \\ 3/2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$2. x^2 + y^2 + z^2 + 2xy - 2xz + 5yz$$

$$3. x^2 + y^2 + z^2 + 3xyz + xy + yz \text{ is not a Q. form.}$$

Matrix of Q.F

$$\text{Let } ax^2 + by^2 + cz^2 + 2hxy + 2gxz + 2fyz = [x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$ax^2 + by^2 + cz^2 + 2hxy + 2gxz + 2fyz$$

$$= [x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Matrix of Q.F is

$$\begin{bmatrix} a & h \\ h & b \end{bmatrix} \text{ or } \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$$

Reduce QF to Canonical form / Sum of squares changing Variables

Step 1: Find the matrix A of QF

Step 2: Find eigen values of A,

Suppose $\lambda_1, \lambda_2, \lambda_3$ are eigen values

$$\text{Matrix of Canonical form is } \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Canonical form is $\lambda_1 s^2 + \lambda_2 t^2 + \lambda_3 v^2$

1. Find canonical form of QF $3x^2 + 6y^2 + 3z^2 - 2yz + 2zx - 2xy$

$$A = \begin{bmatrix} 3 & -1 & +1 \\ -1 & 6 & -1 \\ +1 & -1 & 3 \end{bmatrix} \quad |A| = 3(14) + 1(-4) + 1(14) \\ = 42 - 4 - 4 = 32 \\ = 42 - 2 - 4 = 36$$

$$\lambda^3 - 11\lambda^2 + (14 + 14 + 14)\lambda - 36 = 0$$

$$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

$$\lambda = 2, 3, 6$$

$$\text{Canonical matrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\text{Canonical form is } 2x^2 + 3y^2 + 6z^2$$

2. Find canonical form [Diagonalize] of the QF

$$3x_1^2 + 3x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3$$

Ans: $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix} \quad |A| = 3(9-1) - 1(3+1) + 1(-1-3) \\ = 24 - 4 - 4 = 16$

$$\lambda^3 - 9\lambda^2 + (8+8+8)\lambda - 16 = 0$$

$$\lambda^3 - 9\lambda^2 + 24\lambda - 16 = 0$$

$$\lambda = 1, 4, 0$$

$$\text{canonical matrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\text{Canonical form is } 1s^2 + 4t^2 + 4v^2$$

Assignment Questions

1. Describe nature of roots based on rank of $[A, B]$ and rank of A in the system $Ax = B$.

Ans: 1. $P(A:B) = P(A) = \text{No. of unknowns}$
 $\Rightarrow$ System is consistent and has unique solutions.

2. $P(A:B) = P(A) < \text{No. of unknowns}$
 $\Rightarrow$ System is consistent and has infinitely many solutions.

3. $P(A:B) \neq P(A) \Rightarrow$ System is inconsistent and has no solutions.

2. Find the rank of $A = \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 4 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$

i) By using row reduced echelon canonical form.

ii) By using normal form.

Ans: i) $A = \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 4 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix} \quad R_1 \leftrightarrow R_4$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 4 & 2 & 0 & 2 \\ 4 & -1 & 0 & 3 \\ 2 & -2 & 0 & 6 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_4 \rightarrow R_4 - 2R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 10 & -4 & -6 \\ 0 & 7 & -4 & -5 \\ 0 & 2 & -2 & 2 \end{bmatrix} \sim R_2 \rightarrow R_2/10$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & 7 & -4 & -5 \\ 0 & 2 & -2 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 7R_2$$

$$R_4 \rightarrow R_4 - 2R_2$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & 0 & -6/5 & -2/5 \\ 0 & 0 & -6/5 & 18/5 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_3$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & 0 & -6/5 & -2/5 \\ 0 & 0 & 0 & 18/5 \end{bmatrix}$$

$$R_3 \rightarrow \frac{R_3 \times 5}{6}$$

$$R_4 \rightarrow \frac{R_4 \times 5}{18}$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rank of A = 4

3. Find rank of $A = \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ using Normal form

Ans:

$$A = \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$$

$$R_1 \rightarrow R_1/2$$

$$\sim \begin{bmatrix} 1 & -1 & 0 & 3 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 4R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$\sim \begin{matrix} C_1 & C_2 & C_3 & C_4 \\ \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 6 & 0 & -10 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix} \end{matrix}$$

$$C_2 \rightarrow C_2 + C_1$$

$$C_4 \rightarrow C_4 - 3C_1$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 6 & 0 & -10 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2/6$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -10/6 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix}$$

$$R_4 \rightarrow R_4 + R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -10/6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -16/6 \end{bmatrix}$$

$$C_4 \rightarrow C_4 + \frac{10}{6} C_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -16/6 \end{bmatrix}$$

$$R_3 \leftrightarrow R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -\frac{1}{6} \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad C_4 \rightarrow C_4 + \frac{1}{6} C_3$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \text{order of identity matrix is the rank}$$

$$\therefore \text{Rank} = \underline{\underline{3}}$$

4. Make a change of variable $X = PY$ that transforms the quadratic form $x_1^2 + 10x_1x_2 + x_2^2$ into a quadratic form with no cross product terms. Find P and new quadratic form.

Ans: Matrix of Quadratic form $= \begin{bmatrix} a & h \\ h & b \end{bmatrix}$, Here $a=1, b=1$
 $2h=10, h=5$
 $= \begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$

Characteristic equation is $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 5 \\ 5 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 25 = 0$$

$$1 - 2\lambda + \lambda^2 - 25 = 0$$

$$\lambda^2 - 2\lambda - 24 = 0$$

$$(\lambda - 6)(\lambda + 4) = 0$$

$$\lambda = 6, -4$$

Eigen vector corresponding to $\lambda = 6$

$$[A - 6I]X = 0$$

$$\begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-5x + 5y = 0$$

$$5x - 5y = 0$$

$$5x = 5y$$

$$x = y$$

normalised eigen

$$\text{vector} = \begin{bmatrix} \frac{1}{\sqrt{1+1}} \\ \frac{1}{\sqrt{1+1}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{If } x = C, X = \begin{bmatrix} C \\ C \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Eigen vector corresponding to $\lambda = -4$

$$[A + 4I]X = 0$$

$$\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$5x + 5y = 0$$

$$5x + 5y = 0$$

$$x = -y$$

$$X = \begin{bmatrix} K \\ -K \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{normalised eigen vector} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$P = \begin{bmatrix} \text{Normalised} & \text{normalised} \\ \text{ev}_1 & \text{ev}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$D = P^{-1}AP = \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix}$$

New QF is $Y'DY = \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$
 $= 6y_1^2 - 4y_2^2$

5. Find a and b if the rank of $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & a & b \\ 1 & 1 & -2 & a \end{bmatrix}$ is 2.

Ans: $A = \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & a & b \\ 1 & 1 & -2 & a \end{bmatrix} \quad R_1 \leftrightarrow R_2$

$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 3 & 1 & a & b \\ 1 & 1 & -2 & a \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$

$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 1 & a-3 & b-3 \\ 0 & 1 & -3 & a-1 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array}$

$\sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & a & b-2 \\ 0 & 0 & 0 & a \end{bmatrix} \quad \begin{array}{l} a=0, b-2=0 \\ \therefore a=0 \quad b=\underline{\underline{2}} \end{array}$

6. Reduce quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to the canonical form $Q = Y'DY$

Step 1: Find matrix of QF

Step 2: Find eigen values

Step 3 write $Q = Y' \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} Y = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$

Ans: $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \quad |A| = 3(14) + 1(-2) + 1(-5) = 42 - 2 - 5 = \underline{\underline{36}}$

$\lambda^3 - 11\lambda^2 + (14 + 8 + 14)\lambda - 36 = 0$

$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$

$\lambda = 2, 3, 6$

$Q = Y' \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} Y = 2y_1^2 + \underline{\underline{3y_2^2}} + 6y_3^2$

1. solve $x+y+z=6$
 $x-y+2z=5$
 $3x+y+z=8$

2. solve $x+2y+z=2$
 $3x+y-2z=1$
 $4x-3y-z=3$
 $2x+4y+2z=4$

3. Find Eigen values of $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{bmatrix}$

4. Find eigen values and eigen vectors of $A = \begin{bmatrix} 10 & 3 \\ 4 & 6 \end{bmatrix}$

5. if $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ Find A^2 and A^{-1} using Cayley Hamilton theorem.

6. Using Cayley Hamilton theorem. Find inverse of $A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$

7. Find the rank of $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$

Answers.

Module I

1. $x+y+z=6$

$$x-y+2z=5$$

$$3x+y+z=8$$

we can write this in the form $AX=B$

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \end{bmatrix}$ $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $B = \begin{bmatrix} 6 \\ 5 \\ 8 \end{bmatrix}$

$$A:B = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 3 & 1 & 1 & 8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\sim = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & -2 & -10 \end{bmatrix}$$

$$\begin{array}{c} R_2 \leftrightarrow R_3 \\ \hline R_2 \rightarrow \frac{R_2}{-2} \end{array}$$

$$= \begin{array}{cccc} 1 & 1 & 1 & 6 \\ 0 & 2 & 1 & 1 \end{array}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & -2 & -2 & -10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -3 & -9 \end{bmatrix}$$

$$R_3 \rightarrow R_3 / -3$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

$$\text{Rank of } [A:B] = \text{Rank of } [A] = 3.$$

$\therefore$ System is consistent and has unique values

$$x + y + z = 6 \quad \text{--- (1)}$$

$$y - \frac{1}{2}z = \frac{1}{2}$$

$$x = 6 - 2 - 3 = 6 - 5 = \underline{\underline{1}}$$

$$2y - z = 1 \quad \text{--- (2)}$$

$$z = 3$$

put $z = 3$ in (2).

$$2y = 4$$

$$y = \frac{4}{2} = \underline{\underline{2}}$$

2.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & -2 \\ 4 & -3 & -1 \\ 2 & 4 & 2 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$B = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 4 \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 3 & 1 & -2 & 1 \\ 4 & -3 & -1 & 3 \\ 2 & 4 & 2 & 4 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$R_4 \rightarrow R_4 - 2R_1$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -5 & -5 & -5 \\ 0 & -11 & -5 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 / -5$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & -11 & -5 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 11R_2$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 6 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 / 6$$

$$= \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank}[A:B] = \text{Rank of } A \\ = \underline{\underline{3}}$$

$$x + 2y + z = 2$$

$$x = 2 - 2y - z$$

$$y + z = 1$$

$$x = 2 - 1 = \underline{\underline{1}}$$

$$z = 1$$

$$\therefore y = \underline{\underline{0}}$$

$$3. \quad A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 10 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{bmatrix}$$

$$= (1-\lambda)[(2-\lambda)(3-\lambda)-2] - 1(+2) + 2(1(2-\lambda))$$

$$= (1-\lambda)[6-5\lambda+\lambda^2-2] - 2 + 4 - 2\lambda$$

$$= 1 - 6\lambda + 5\lambda^2 - \lambda^3 + 2\lambda - 2 + 4 - 2\lambda$$

$$= -\lambda^3 + 5\lambda^2 - 6\lambda + 3 = 0$$

$$\lambda^3 - 5\lambda^2 + 6\lambda = 3$$

$$= (1-\lambda)[4-5\lambda+\lambda^2] + 2 + 4 - 2\lambda$$

$$= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda = 6$$

$$\lambda(\lambda^2 - 6\lambda + 11) =$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\lambda = 1, \lambda = 3, \lambda = 2$$

$$\lambda = \underline{\underline{1, 2, 3}}$$

$$A = \begin{bmatrix} 10 & 3 \\ 4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 3 \\ 4 & 6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 10-\lambda & 3 \\ 4 & 6-\lambda \end{bmatrix}$$

$$\Delta (10-\lambda)(6-\lambda) - 12 = 0$$

$$60 - 16\lambda + \lambda^2 - 12 = 0$$

$$\lambda^2 - 16\lambda + 48 = 0$$

Sum

$$\left. \begin{array}{l} \text{sum} = -16 \\ \text{pdt} = 48 \end{array} \right\} -12, -4$$

$$(\lambda - 4)(\lambda - 12) = 0$$

$$\lambda = 12, 4$$

Eigen vector with eigen value 12

$$[A - 12I]x = 0$$

$$\begin{bmatrix} 10 & 3 \\ 4 & 6 \end{bmatrix} - \begin{bmatrix} 12 & 0 \\ 0 & 12 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 3 \\ 4 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2x + 3y = 0$$

$$4x - 6y = 0$$

$$2x = 3y$$

$$x = \frac{3y}{2}$$

$$\text{put } y = k$$

$$\therefore x = \frac{3k}{2}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{3k}{2} \\ k \end{bmatrix}$$

$$\text{where } k = 1$$

$$= \begin{bmatrix} 3/2 \\ 1 \end{bmatrix}$$

eigen vector with eigen value 4

$$[A - 4I]x = 0$$

$$\begin{bmatrix} 6 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$6x + 3y = 0$$

$$4x + 2y = 0$$

$$4x = -2y$$

$$x = -\frac{y}{2}$$

$$\text{Let } x = c$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \\ -2c \end{bmatrix} = c \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$5. A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) + 2$$

$$= 3 - 4\lambda + \lambda^2 + 2$$

$$= \lambda^2 - 4\lambda + 5 = 0$$

$$\lambda^2 - 4\lambda + 5 = 0$$

~~sum~~ = 4

~~pdt~~ = 5

So,

$$A^2 - 4A + 5I = 0$$

$$A^2 - 5I + 4A = 4A - 5I = 4 \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 8 \\ -4 & 12 \end{bmatrix} - \begin{bmatrix} 6 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix}$$

$$A^2 - 4A + 5I = 0$$

xy by A^{-1} ,

$$A^2 A^{-1} - 4A A^{-1} + 5I A^{-1} = 0$$

$$A - 4I + 5A^{-1} = 0$$

$$5A^{-1} = 4I - A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3/5 & -2/5 \\ 1/5 & 1/5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$$

$$\text{Ch eqn} \Rightarrow |A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & -2 \\ 2 & -1-\lambda & 4 \\ 3 & 1 & -1-\lambda \end{vmatrix} = 1-\lambda \left[(-1-\lambda)(-1-\lambda)-4 \right] - 2(2(1+\lambda)-12) +$$

$$-2(2+3(1+\lambda)) = 0$$

$$= (1-\lambda)(1+2\lambda+\lambda^2-4) - 2(-2-2\lambda-12) - 2(2+3+3\lambda) = 0$$

$$= (1-\lambda)(\lambda^2+2\lambda-3) + 28+4\lambda-10-6\lambda = 0$$

$$= 2\lambda - 2\lambda^2 + \lambda^2 - \lambda^3 - 3 + 3\lambda + 28 + 4\lambda - 10 - 6\lambda$$

$$= -\lambda^3 - \lambda^2 + 3\lambda - 15 = 0 \Rightarrow \lambda^3 + \lambda^2 - 3\lambda - 15 = 0$$

$$A^3 + A^2 - 3\lambda - 15I = 0$$

$$A^3 A^{-1} + A^2 A^{-1} - 3\lambda A^{-1} - 15I A^{-1} = 0$$

$$A^2 + A - 3I - 15A^{-1} = 0$$

$$A^2 = \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4-6 & 2-2-2 & -2+8+2 \\ 2+2+12 & 4+1+4 & -4-4-4 \\ 3+2-3 & 6-1-1 & -6+4+1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -2 & 8 \\ 12 & 9 & -12 \\ 2 & 4 & -1 \end{bmatrix}$$

$$15A^{-1} = A^2 + A - 3I$$

$$= \begin{bmatrix} -1 & -2 & 8 \\ 12 & 9 & -12 \\ 2 & 4 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 2 & -2 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 0 & 6 \\ 14 & 5 & -8 \\ 5 & 5 & -5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -\frac{3}{15} & 0 & \frac{6}{15} \\ \frac{14}{15} & \frac{5}{15} & -\frac{8}{15} \\ \frac{5}{15} & \frac{5}{15} & -\frac{5}{15} \end{bmatrix} = \begin{bmatrix} -1/5 & 0 & 2/5 \\ 14/15 & 1/3 & -8/15 \\ 1/3 & 1/3 & -1/3 \end{bmatrix}$$

7.

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$R_4 \rightarrow R_4 - 6R_1$$

$$= \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -8 & 3 \\ 0 & -4 & -11 & 5 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$= \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & -4 & -11 & 5 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - R_2$$

$$= \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & -4 & -8 & 3 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 / -4$$

$$R_3 \rightarrow R_3 / -3$$

$$= \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -3/4 \\ 0 & 0 & 1 & -2/3 \\ 0 & 0 & -3 & 2 \end{bmatrix}$$

$$R_4 = R_4 + 3R_3$$

$$= \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & -3/4 \\ 0 & 0 & 1 & -2/3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank of } [A:B] = \text{Rank of } [A] = \underline{\underline{3}}$$

02/04/24

Class Test

Module 1

1. Find the rank of $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{bmatrix}$

Ans: $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{bmatrix}$
 $R_2 \rightarrow R_2 - 2R_1$
 $R_3 \rightarrow R_3 - 3R_1$
 $R_4 \rightarrow R_4 - 4R_1$

$\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\therefore \text{Rank of } A = \underline{\underline{1}}$

2. Find the characteristic Equ of $A = \begin{bmatrix} 5 & 10 \\ 2 & 5 \end{bmatrix}$

char equ is $|A - \lambda I| = 0$

$\begin{vmatrix} 5-\lambda & 10 \\ 2 & 5-\lambda \end{vmatrix} = 0$

$(5-\lambda)^2 - 20 = 0$

$25 - 10\lambda + \lambda^2 - 20 = 0$

$\lambda^2 - 10\lambda + 5 = 0$

3. Solve the system of eq $2x+3y-3z=1$, $4x+y-4z=5$, $3x-y-2z=6$

Ans:

$2x+3y-3z=1$

$4x+y-4z=5$

$3x-y-2z=6$

Ans

$A:B = \begin{bmatrix} 2 & 3 & -3 & 1 \\ 4 & 1 & -4 & 5 \\ 3 & -1 & -2 & 6 \end{bmatrix}$

$R_1 \rightarrow R_1/2$

$\sim \begin{bmatrix} 1 & 3/2 & -3/2 & 1/2 \\ 4 & 1 & -4 & 5 \\ 3 & -1 & -2 & 6 \end{bmatrix}$

$R_2 \rightarrow R_2 - 4R_1$

$R_3 \rightarrow R_3 - 3R_1$

$\sim \begin{bmatrix} 1 & 3/2 & -3/2 & 1/2 \\ 0 & -5 & 2 & 3 \\ 0 & -11/2 & 5/2 & 9/2 \end{bmatrix}$

$R_2 \rightarrow R_2/-5$

$\sim \begin{bmatrix} 1 & 3/2 & -3/2 & 1/2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & -11/2 & 5/2 & 9/2 \end{bmatrix}$

$R_3 \rightarrow R_3 + \frac{11}{2}R_2$

$\sim \begin{bmatrix} 1 & 3/2 & -3/2 & 1/2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & 0 & 3/10 & 6/5 \end{bmatrix}$

$R_3 \rightarrow \frac{10}{3}R_3$

$\sim \begin{bmatrix} 1 & 3/2 & -3/2 & 1/2 \\ 0 & 1 & -2/5 & -3/5 \\ 0 & 0 & 1 & 4 \end{bmatrix}$

$\text{Rank}[A] = \text{Rank}[A:B] = 3$

$x + \frac{3}{2}y - \frac{3}{2}z = \frac{1}{2}$

$y - \frac{2}{5}z = -\frac{3}{5}$

$z = \underline{\underline{4}}$

$y = \frac{2}{5}z - \frac{3}{5} = \frac{8}{5} - \frac{3}{5} = \frac{5}{5} = \underline{\underline{1}}$

$x = \frac{3}{2}z + \frac{1}{2} - \frac{3}{2}y = 6 + \frac{1}{2} - \frac{3}{2} = 6 - 1 = \underline{\underline{5}}$

4. Find the eigen values and eigen vectors [basis for null vectors

$$\text{of } A - \lambda I] \text{ of } A = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

Ans: char equ $\lambda^3 - 3\lambda^2 + (-3 - 3 - 3)\lambda - 5 = 0$

$$\lambda^3 - 3\lambda^2 - 9\lambda - 5 = 0$$

$$\lambda = 5, -1, -1$$

Eigen vectors corresponding to eigen value $\lambda = 5$,
 $[A - 5I]X = 0$

$$\begin{bmatrix} -4 & -2 & 2 \\ -2 & -4 & -2 \\ 2 & -2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-4x - 2y + 2z = 0 \Rightarrow -2x - y + z = 0 \quad \text{--- (1)}$$

$$-2x - 4y - 2z = 0 \Rightarrow -x - 2y - z = 0 \quad \text{--- (2)}$$

$$2x - 2y - 4z = 0 \Rightarrow x - y - 2z = 0 \quad \text{--- (3)}$$

$$\text{(1) + (2)} \quad -3x - 2y = 0$$

$$\therefore x = -y$$

$$\text{thus in (2), } -y - z = 0$$

$$y = -z$$

$$x = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

5. Find a and b if the rank of $\begin{bmatrix} 2 & 3 & 4 & 5 \\ 1 & 2 & a & b \\ 6 & 9 & 9+a & 10+b \end{bmatrix}$ is 2

Ans:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 1 & 2 & a & b \\ 6 & 9 & 9+a & 10+b \end{bmatrix} \quad R_1 \rightarrow R_1/2$$

$$\sim \begin{bmatrix} 1 & 3/2 & 2 & 5/2 \\ 1 & 2 & a & b \\ 6 & 9 & 9+a & 10+b \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 6R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 3/2 & 2 & 5/2 \\ 0 & 1/2 & a-2 & b-\frac{5}{2} \\ 0 & 0 & -3+a & -5+b \end{bmatrix}$$

$$-3+a=0 \Rightarrow a = \underline{\underline{3}}$$

$$-5+b=0 \Rightarrow b = \underline{\underline{5}}$$

6. Using Cayley Hamilton thm Find the inverse of $A = \begin{bmatrix} 2 & 3 & -8 \\ 4 & 1 & -4 \\ 3 & -1 & -2 \end{bmatrix}$

Ans: char eqn is

$$\lambda^3 - \lambda^2 + (-6+5-10)\lambda + 3 = 0$$

$$\lambda^3 - \lambda^2 - 11\lambda + 3 = 0$$

$$A^3 - A^2 - 11A + 3I = 0$$

$$A^3 A^{-1} - A^2 A^{-1} - 11A A^{-1} + 3I A^{-1} = 0$$

$$A^2 - A - 11I + 3A^{-1} = 0$$

$$A^2 = \begin{bmatrix} 2 & 3 & -3 \\ 4 & 1 & -4 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 3 & -3 \\ 4 & 1 & -4 \\ 3 & -1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+12-9 & 6+3+3 & -6-12+6 \\ 8+4-12 & 12+1+4 & -12-4+8 \\ 6-4-6 & 9-1+2 & -9+4+4 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 12 & -12 \\ 0 & 17 & -8 \\ -4 & 10 & -1 \end{bmatrix}$$

$$3A^{-1} = A + 11I - A^2$$

$$= \begin{bmatrix} 2 & 3 & -3 \\ 4 & 1 & -4 \\ 3 & -1 & -2 \end{bmatrix} + \begin{bmatrix} 11 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 11 \end{bmatrix} - \begin{bmatrix} 7 & 12 & -12 \\ 0 & 17 & -8 \\ -4 & 10 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -9 & 9 \\ 4 & -5 & 4 \\ 7 & -11 & 10 \end{bmatrix} \quad A^{-1} = \frac{1}{3} \begin{bmatrix} 6 & -9 & 9 \\ 4 & -5 & 4 \\ 7 & -11 & 10 \end{bmatrix}$$

7. Find the canonical form (sum of square form) of quadratic form $6x^2 + 3y^2 + 3z^2 - 4xy - 24z + 4xz$

Ans: $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ $|A| = 6(8) + 2(-4) + 2(-4) = 48 - 8 - 8 = 32$

$$\lambda^3 - 12\lambda^2 + (8+14+14)\lambda - 32 = 0$$

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$$\lambda = 8, 2, 2$$

$$\text{Canonical matrix} = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\text{Canonical form} = 8x^2 + 2y^2 + 2z^2$$